

RNN Text Classifier — Toxic Category Detection

Bidirectional LSTM trained on cellula_toxic_data.csv

1. Task Overview

The dataset contains 3,000 samples with a text 'query', an 'image descriptions' field, and a target label 'Toxic Category' with 9 classes: Child Sexual Exploitation, Elections, Non-Violent Crimes, Safe, Sex-Related Crimes, Suicide & Self-Harm, Unknown S-Type, Violent Crimes, unsafe. The query and image description were concatenated into a single text field and used to train a Recurrent Neural Network (Bidirectional LSTM) to classify each sample into its correct toxic category.

Split	Samples
Train	2100
Validation	450
Test	450

2. Model Architecture

Embedding(vocab=10,000, dim=128) → SpatialDropout1D(0.3) → Bidirectional LSTM(64, return_sequences=True) → Bidirectional LSTM(32) → Dense(64, ReLU) → Dropout(0.4) → Dense(9, Softmax). Trained with Adam (lr=1e-3), categorical cross-entropy loss, class weighting to counter class imbalance, and early stopping on validation loss.

3. Evaluation Results (Held-out Test Set)

Metric	Score
Accuracy	0.916
F1 Score (Weighted)	0.924
F1 Score (Macro)	0.929
Target F1	0.850

Result: F1 (weighted) = 0.924 ≥ target of 0.85 → PASSED

4. Per-Class Classification Report

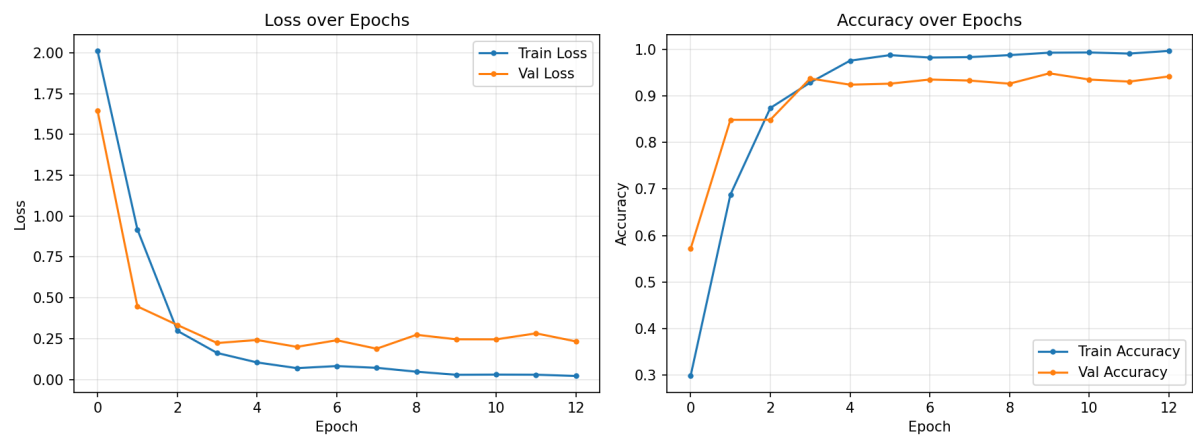
precision recall f1-score support

Child Sexual Exploitation 1.000 1.000 1.000 15
Elections 1.000 1.000 1.000 17
Non-Violent Crimes 1.000 1.000 1.000 45
Safe 0.938 0.813 0.871 150
Sex-Related Crimes 1.000 0.941 0.970 17
Suicide & Self-Harm 1.000 1.000 1.000 17
Unknown S-Type 0.412 0.724 0.525 29
Violent Crimes 1.000 0.992 0.996 119
unsafe 1.000 1.000 1.000 41

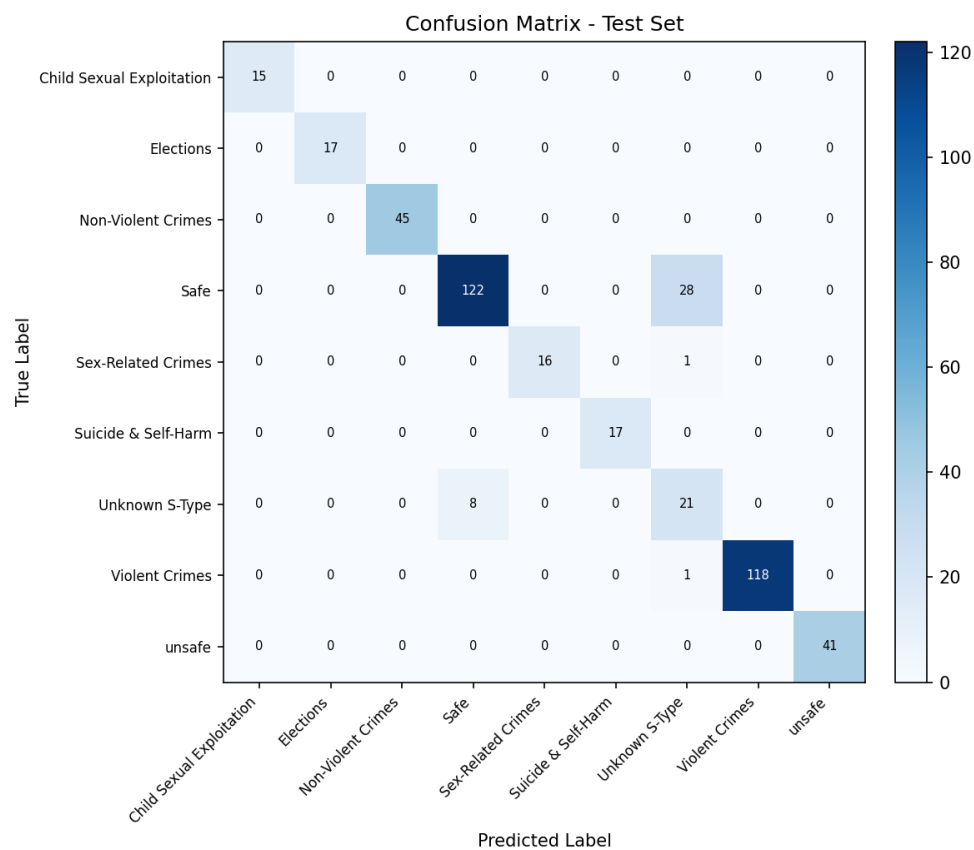
accuracy 0.916 450

macro avg 0.928 0.941 0.929 450
weighted avg 0.942 0.916 0.924 450

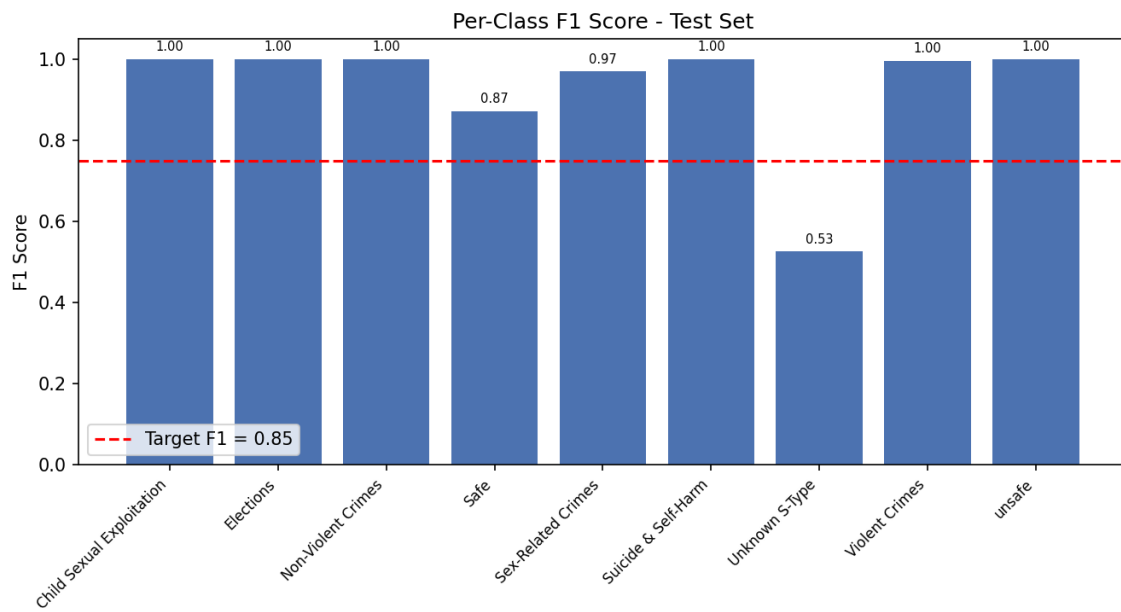
5. Training Curves



6. Confusion Matrix



7. Per-Class F1 Score



8. Conclusion

The Bidirectional LSTM model achieved a weighted F1 score of 0.924 and a macro F1 score of 0.929 on the held-out test set, exceeding the required threshold of 0.85.